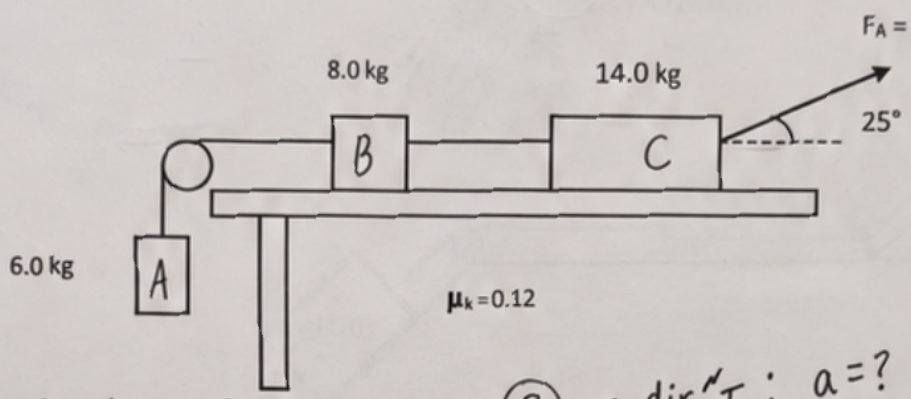
2.5 Complex Systems

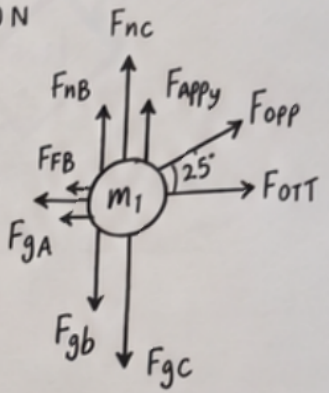
Learning Goals (Students will be able to….)

- Solve complex problems involving inclined planes and pulleys combined.

Putting It All Together

1. A string and pulley system is set up as shown below. The string is massless and inextensible and the pulley frictionless. The coefficient of kinetic friction between the 8.0 kg and 14.0 kg blocks with the table is 0.12. Calculate:
   a) the acceleration of the 8.0 kg block [0.39 m/s^2]
   b) the tension in the string between the 8.0 kg and 14.0 kg block. [74 N]





① y-dir ^ B : F_{nB} = ?

[I] $F_{netyB} = F_{nB} - F_{gB}$

[II] $F_{netyB} = m_B a_y$

[II] − [2] → $0 = F_{nB} - F_{gB}$

$F_{nB} = m_B g$

② y-dir ^ C : F_{nC} = ?

[I] $F_{netyC} = F_{nC} + F_{appy} - F_{gC}$

[2] $F_{netyC} = m_C a_y$

[II] − [2] → $0 = F_{nC} + F_{app}\sin\theta - m_C g$

$F_{nC} = m_C g - F_{app}\sin\theta$

③ x-dir ^ T : a = ?

[I] $F_{netTx} = F_{appx} - F_{fB} - F_{fC} - F_{gA}$

[II] $F_{netTx} = m_T a$

[II] − [2] → $m_T a = F_{appx} - F_{fB} - F_{fC} - F_{gA}$

$m_T a = F_{app}\cos\theta - \mu F_{nB} - \mu F_{nC} - m_A g$

$a = \dfrac{F_{app}\cos\theta - \mu(m_C g - F_{app}\sin\theta + m_B g) - m_A g}{m_A + m_B + m_C}$

$a = 0.39$ N/kg

$\therefore \ddot{a} = 0.39$ N/kg [x-dir$^+$]

b) x-dir ^ C : F_{BonC} = ?

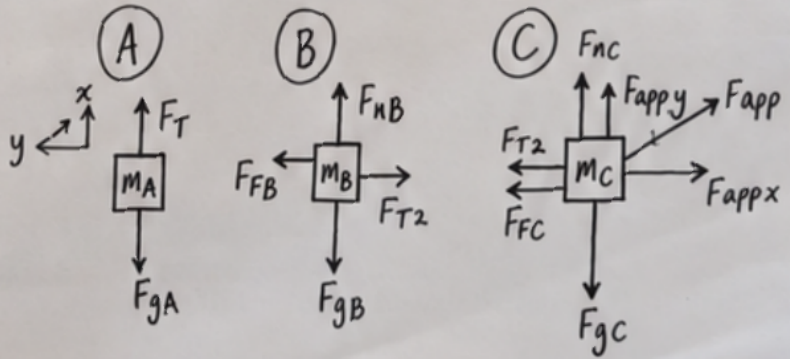
[I] $F_{netxC} = F_{appx} - F_{fC} - F_T$

[2] $F_{netxC} = m_C a$

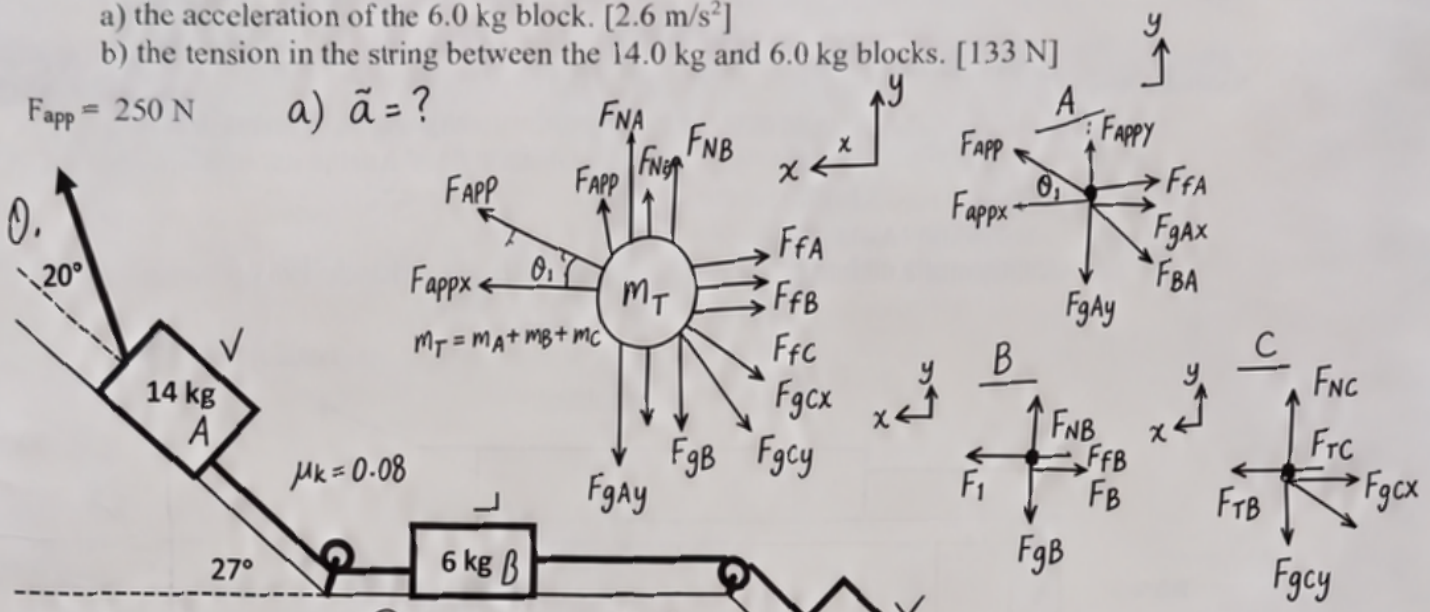
[II] − [2] → $m_C a = F_{app}\cos\theta - \mu(m_C g - F_{app}\sin\theta) - F_T$

$F_T = F_{app}\cos\theta - \mu(m_C g - F_{app}\sin\theta) - m_C a$

$F_T \cong 74$ N



2) A string and pulley system is set up as shown below. The string is massless and inextensible and the pulley frictionless. The coefficient of kinetic friction between the blocks and the surface is 0.08. Calculate:
a) the acceleration of the 6.0 kg block. [2.6 m/s^2]
b) the tension in the string between the 14.0 kg and 6.0 kg blocks. [133 N]



① y-dir^m A : F_{NA} = ?

☐ $F_{net,Ay} = F_{app.y} + F_{NA} + F_{gAy}$
☒ $F_{net,Ay} = m_A a_y = 0$
☐ $-0 = F_{app.y} + F_{NA} - F_{gAy}$
$F_{NA} = m_A g\cos\theta_2 - F_{app}\sin\theta_1$

② y-dir^m B : F_{NB} = ?

☐ $F_{net.y} = F_{NB} - F_{gB}$
☒ $F_{net.y} = m_B a_y = 0$
☐ $-0 = F_{NB} - F_{gB}$
$F_{NB} = m_B g$

③ y-dir^m C : F_{NC} = ?

☐ $F_{net.y} = F_{NC} - F_{gCy}$
☒ $F_{net.y} = m_C a_y = 0$
☐ − ☒ → $0 = F_{NC} - F_{gCy}$
$F_{NC} = m_C g\cos\theta_3$

④ x-dir^m sys: $\tilde{a}$ = ?

☐ $F_{net,Tx} = F_{appx} - F_{fA} - F_{fB} - F_{fC} - F_{gAx} - F_{gCx}$
☒ $F_{net,Tx} = m_T a$
☐ − ☒
$m_T a = F_{appx} - F_{fA} - F_{fB} - F_{fC} - F_{gAx} - F_{gCx}$
$m_T a = F_{app}\cos\theta_1 - \mu_k F_{NA} - \mu_k F_{NB} - \mu_k F_{NC} - m_A g\sin\theta_2 - m_C g\sin\theta_3$

$$a = \frac{F_{app}\cos\theta_1 - \mu_k(m_A g\cos\theta_2 - F_{app}\sin\theta_1 + m_B g + m_C g\cos\theta_3) - g(m_A\sin\theta_2 + m_C\sin\theta_3)}{m_A + m_B + m_C}$$

$a \approx 2.62$

$\therefore \tilde{a} \approx 2.6$ m/s^2 [x-dir$^+$]

b) $\tilde{F}_{BonA}$ = ?

① x-dir^m A : F_T = ?

☐ $F_{net,Ax} = F_{appx} - F_{fA} - F_{gAx} - F_{BonA}$
☐ $F_{net,Ax} = m_A a$
☐ $- m_A a = F_{appx} - F_{fA} - F_{gAx} - F_{BonA}$
$F_{BonA} = F_{app}\cos\theta_1 - \mu_k(m_A g\cos\theta_2 - F_{app}\sin\theta_1) - m_A g\sin\theta_2 - m_A a$
$F_{BonA} \approx 133$ N
$\therefore F_{BonA} \approx 133$ N

Homework

Finish the classwork problems (if not done).